

Producing radioactive diamonds

Natural diamonds, made up entirely of carbon-12, take hundreds of thousands to millions of years to form naturally when subject to high pressures deep in the earth's crust. These do not contain any radioactive carbon. Artificial diamonds are produced using a good carbon source such as graphite and a machine to create high pressure. However, these do not contain any radioactive material. Radioactive man-made diamonds for usage in diamond batteries are made from radioactive C-14 isotopes, which are known to be the largest nuclear waste by volume.

In nuclear power reactors, fissile uranium rods are placed inside graphite blocks that act as neutron moderators to slow down the fast-moving neutrons. During the nuclear fission, uranium atoms get split, generating heat that vaporises water into steam, which turns electricity-generating turbines.

During the process, some of the non-radioactive C-12 and C-13 isotopes in graphite blocks capture neutrons and get converted into radioactive C-14, which mainly concentrates on the outer parts of the blocks and can be effectively removed by heating them to the sublimation point of 3,642 degrees centigrade.

This results in releasing C-14 in gaseous form, and disposal of the less radioactive graphite blocks also becomes easier. C-14 gas is collected and subject to low pressure and elevated temperature to produce artificial radioactive diamonds using a process called chemical vapour deposition (CVD).

The diamonds are encased in non-radioactive artificial diamonds that provide protection by containing the radiation. It is highly dangerous to ingest or touch radioactive diamonds with naked skin, but they remain safe when encapsulated as no short-range radiation can escape. Being the hardest substance known, diamonds offer the best protection. Scott and his team have managed to create a prototype using a blend of C-14 and bits of tritium (a form of radioactive hydrogen) as the source of beta-radiation.

How diamond batteries work

Functionally, the concept of a diamond battery is similar to that of radioisotope electric generators used for space probes and rovers. The diamond battery generates electric charge through direct energy conversion using a semiconductor diode and C-14 as a radioisotope source. C-14, while undergoing decay, emits short-range low-energy beta particles (essentially the nucleus' version of electrons) and turns into nitrogen-14, which is not radioactive and gets absorbed in the diamond casing.

The beta particles released by C-14, moving with an average energy of 50keV, undergo inelastic collisions with other carbon atoms present in the diamond structure, causing electrons to jump from the valence band to the conduction band, leaving behind holes in the valence band where electrons were earlier present. Successive electron-hole pairs get collected at the metal contact attached to the diamond. C-14 is preferred as the source material because its beta particle radiation is easily absorbed by any solid.

Some devices use tritium and beryllium along with C-14 as additional beta-emitting radioisotopes. The amount of C-14 used in each battery is yet to be disclosed, but what is known is that a battery containing 1gm of C-14 would deliver 15 joules (15J) per day, much less than an AA battery, and would take 5,730 years to reach fifty per cent power. Alkaline AA batteries are designed for short timeframe discharge, and one battery weighing about 20gm has an energy storage rating of 700J/gm. When operated continuously, it would run out in 24 hours.

Diamond batteries do not provide power directly to attached devices but are designed with a capacitor that gets 'trickle charged' by the battery and then discharges to power the devices at set intervals. The following characteristics are common for a prototype:

- Voltage: 2V (estimated)
- Energy: 31MJ over first 5,700 years
- Prototype size: 10mm × 10mm × 0.5mm (plus electrodes)
- Temperature: Physically stable at 750 degrees Celsius



Graphite blocks in a nuclear reactor
(Source: www.popularmechanics.com)

Powering the future-nuclear batteries

Atomic battery, nuclear battery, tritium battery, or radioisotope generator are devices that use energy from the decay of radioactive isotopes to generate electric current. These differ from nuclear reactors and do not use a chain reaction. Nuclear batteries use radioactivity to generate power instead of storing a certain amount of charge.

Compared to chemical batteries, nuclear batteries are characterised by higher volumetric energy density (therefore longer battery life) and stronger endurance in harsh conditions. Nuclear batteries are used in space exploration applications. **It is impossible or extremely costly to access a device once it has been launched into space, and solar energy is not a practical solution to powering electronic equipment since its availability is limited.**

NASA uses nuclear battery technology called Radioactive Thermoelectric Generator (RTG) to power spacecraft. Batteries used for implantable medical devices (IMDs) need to function over a long period of time without being accessed for recharge or maintenance and should be limited in size and weight. Nuclear batteries called betavoltaic cells were widely